

8. RED-TEAMING: A proactive method of identifying vulnerabilities by simulating attacks on an AI system, organisation or network

9. ZERO-DAY VULNERABILITY: A previously unknown security flaw in software or hardware, giving its developers 'zero days' to patch it before it gets exploited

10. SOVEREIGN AI: AI systems and the underlying infrastructure developed and operated within a country, giving it greater control over the technology and its data

who has been actively involved in framing AI governance policy in India, points out that the country's dependence on imported technology creates clear risks, from supply-chain pressure or coercion, hidden backdoors or trojans in hardware and software, whether from adversarial or even friendly countries, to loss of operational freedom if core AI capabilities—models, computing power or updates—get disrupted, spoofed or switched off through a remote kill switch. "Cyberattacks, particularly on critical infrastructure, would disrupt capability at national scale. These could spill into the defence arena if attacks target AI model weights, data relays and compute infrastructure such as data centres, undersea cables, satellites, etc. Adversaries can disrupt power and cooling for AI data centres, or sabotage network links to degrade battlefield AI," says Anand, pointing to the recent Iranian attacks on data centres in Qatar, Bahrain and the UAE. Worse, AI systems themselves can be weaponised to launch faster, larger-scale cyberattacks on critical infrastructure at machine speed.



"Cyberattacks on critical infrastructure could spill into defence, **targeting model weights, training pipelines, data centres**, undersea cables and satellites"

MAJ. GEN. PAWAN ANAND (RETD), *Director, Centre for Emerging Technologies for Atma Nirbhar Bharat*

Hence, he says, we need a partially if not fully sovereign AI stack for critical military applications, including command and control, strategic intelligence, surveillance and reconnaissance, autonomous weapons and cyber defence. There is already existing capability; it must now be coordinated across the relevant ministries and the military. "Strategic autonomy," adds Anand, "does not mean isolation. India speaks of *aatmanirbharta*, which is self-reliance, not self-sufficiency; it should work with countries with overlapping interests and complementary strengths."

The Infra Danger

Indian Railways is planning to double traffic by using AI for optimisation and scheduling, with predictive maintenance an added advantage, says Anshul Gupta, advisor, RailTel and AI System Development Facilitator. The critical issue, he points out, is to think through the technology upgrade itself. If agentic AI is an immediate concern, the need to migrate to quantum-safe networks is already on the horizon. "Our infrastructure is highly vulnerable. Number one, they are legacy systems; number two, we keep making incremental improvements. It is harmful if you put AI into any equipment without thinking it through," Gupta says.

"Today, digital technologies, including AI, have become pervasive and critical operational foundations of any airline," says Dr Satya Ramaswamy, Air India's chief digital and technology officer. The airline has modernised more than 140 IT and digital systems, spanning reservations, departure control, crew and maintenance planning, safety, compliance and disruption management. It also has more than 100 AI initiatives and 30 agent-led workflows under way.

Infrastructure major Larsen & Toubro uses agentic AI for project management, cost control, supply chains and financial processes. "These agents recommend actions, and also bring autonomy into the workflow," says R. Ganesan, senior vice president and head-corporate, L&T, adding that they can coordinate across systems and trigger approvals with minimal human intervention. The company also deploys AI agents in engineering software, along with computer vision and spatial AI at construction sites, creating what Ganesan calls a 'physical AI' layer, from autonomous painting and plastering to material movement, progress monitoring and detecting safety and quality gaps.

All this means, as Vinayak Godse, CEO of the Data Security Council of India (DSCI), says, that "it's a new world for cybersecurity". "Until now, our IT systems behaved in a particular way...we had a small number of actions to control. Now, we are talking to AI systems in a million ways. So, we have to be mindful of the scale, complexity and interdepen-